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Question	Answer	Marks
1	**Defining the problem**	
	d is the independent variable and *e* is the dependent variable, or vary *d* and measure *e.*	**1**
	Keep *F* constant.	**1**
	Methods of data collection	
	Labelled workable diagram including elastic cord fixed at one end to a support and load attached to the other end Cord and weight must be labelled	**1**
	Use of ruler to measure unstretched length and stretched length. or labelled ruler drawn parallel to cord and original length *L* and either *e* or stretched length indicated.	**1**
	Use of a micrometer / (calipers) to determine *d*.	**1**
	Weigh load on a balance or use of balance to measure mass of load and multiply by *g*	**1**
	Method of Analysis	
	Plots a graph of *e* against $1 / d^2$ or equivalent	**1**
	Relationship valid if a straight line passing through the origin is produced	**1**
	$E = \frac{4FL}{gradient \times \pi}$	**1**
	Additional detail including safety considerations	**Max 6**
	Use safety goggles / safety screen to prevent injury to eyes from (moving) elastic cord / load or use cushion / sand box in case load falls.	**D1**
	Keep *L* constant	**D2**
	Method to keep *L* constant, e.g. check length of each cord / adjust through cork	**D3**

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Question	Answer	Marks
1	Additional detail on measuring *e*, e.g. record initial position, record final position and subtract	**D4**
	Repeat measurement of *d* along cord / different diameters and average	**D5**
	Method to ensure Hooke's law is obeyed, e.g. check that the length is constant after removing load or do not exceed elastic limit	**D6**
	Wait for cord to extend to its maximum value / stop oscillating	**D7**
	Use of $A = \frac{\pi d^2}{4}$	**D8**
	Use of set square to check that ruler is vertical to bench, or use of set square as a fiducial mark to read measurements	**D9**

Question	Answer	Marks
2(a)	Gradient = $\frac{D}{S}$	**1**
2(b)	3.95 4.35 4.80 5.20 5.55 5.90	**1**
	Absolute uncertainties in $P \pm 0.05$	**1**